

① <MPCA - Sample Questions for Practice>

- ① What do you understand by term 'Microprocessor'? Highlight the brief history of generation of microprocessors.
 - ② Differentiate between microcontrollers and microprocessors.
 - ③ What are the main components of microprocessors? Mention the application areas of microprocessor and microcontroller.
 - ④ Block diagram for both computer architecture and computer organisation & explanation.
 - ⑤ Differentiate between Von Neumann Architecture and Harvard Architecture.
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- ⑥ What are the functional units of 8085 Microprocessor? Explain with block diagram of 8085 microprocessor.
 - ⑦ Explain the instruction classification and flow of an instruction in 8085 microprocessor.
 - ⑧ Explain the flag register of 8085 microprocessor.
 - ⑨ What are timing and control unit operations which comes under 8085 microprocessor?
 - ⑩ Explain the pin diagram of 8085 microprocessor.
 - ⑪ What is addressing modes? Explain the types of addressing modes available in 8085 microprocessor with examples.
 - ⑫ Why do you need instruction cycle in 8085 microprocessor? Also define t-states, timing diagram, machine cycle briefly.
 - ⑬ What is timing diagram? Design a timing diagram of STA 9000H.
 - ⑭ Timing diagram for MOV A, B
 - ⑮ Timing diagram for MVI B, 43
 - ⑯ Timing diagram for opcode fetch cycle for MOV C, A.
[You can research more examples of timing diagram by yourself]

- ⑭ Write an ALP of 8086 to add two 8 bit numbers 01H and 05H and store the result at memory location 2050H.
 - ⑮ Write an ALP to subtract two 8 bit numbers 05 and 01H and store the result at memory location 2050H.
 - ⑯ Write an ALP to add two 8 bit numbers 7FH and 9AH with carry.
 - ⑰ Write an ALP program to multiplication of two 8 bit numbers 04H and 03H.
 - ⑱ Write an ALP in 8085 microprocessor to find square of a 8 bit number.
 - ⑳ See also the programs which you have done in class along with others programs to show factorial, reversing string, sum of array.
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- ㉑ Explain the architecture of 8086 microprocessor with neat block diagram.
 - ㉒ Explain the addressing modes of 8086 microprocessor.
 - ㉓ What are ALP development tools? Why do you need them.
 - ㉔ What is the role of flag register in 8086 microprocessor?
 - ㉕ Differentiate between 8085 and 8086 microprocessor.
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- ㉖ You will be asked any one CPU designing question. So refer my class lecture, my pdf and past question from COA. It will be not so logical to write all those questions here. You can refer youtube.com too. Be sure to practice it lot.
 - ㉗ Designing a hardwired control unit and control signal generation, ALU & Register Section.
 - ㉘ Designing a horizontal and vertical microcode for the same question which will be asked.
 - ㉙ Reducing the numbers of microinstructions.

- (32) Differentiate between microprogrammed vs Hardwired control.
- (33) How do you represent number in computer? Represent -9 in 8 bit format. Explain also the 1's complement and 2's complement with examples.
- (34) Design a flow chart for Addition and Subtraction of ~~signed~~ number. Use it to subtract (6-5). [See other examples too].
- (35) Write the RTL code of shift add multiplication algorithm. Use it to multiply 13×11 . [See more such examples].
- (36) Write RTL code for Booth's Algorithm. Perform $(-3) \times (-5)$, $9 \times (-3)$, $(-7) \times (-2)$, $(-9) \times (-12)$ etc.... using Booth's algorithm. Any one such type of question will be asked.
- (37) What is floating point number? Explain IEEE 754 floating point number representation. Represent $(1259.125)_{10}$ in single and double precision format.
- (38) Practise numerical which we have done in class. Conversion of floating point binary number into decimal.
- (39) Conversion decimal to binary. [See more too].
- (40) What is Look up ROM? Explain with an example how it works.
- (41) What is arithmetic pipeline? How do you add or sub two floating numbers using arithmetic pipeline? Give an example.
- (42) Explain Wallace tree with suitable and understandable example.
- (43) Explain memory hierarchy. Explain with block diagram the Associative memory.
- (44) What is cache memory? Explain the cache mapping techniques with example.
- (45) What is cache memory? Explain the working mechanism of direct mapping.

- (46) Differentiate fully associative mapping and set associative mapping. Consider a direct mapped cache of size 32 kB with block size 32 bytes. The CPU generates 32 bit address. The no. of bits required for cache indexing & tag bits respectively. ^{Find}
- (47) Explain Cache Replacement Algorithms with suitable examples of each. [See class notes & numerals]. (3)
- (48) Differentiate between serial & parallel Interface / Communication.
- (49) What are different types of I/O techniques. Explain.
- (50) Why do you need DMA? Explain working process of DMA with diagram.
- (51) Why do we need input output module? Compare programmed I/O, Interrupt driven data transfer and Direct memory Access.
- (52) USB and HDMI Standard + UART. (9)
- (53) Differentiate between RISC vs CISC standards.
- (54) Explain instruction pipeline. In a system, there are 5 windows of registers, each registers share 4 input registers and 4 output registers. Each of window has 10 local registers. The system has 20 global registers. Calculate the total no. of registers in the system. Show pictorial representation as well.
- (55) What is Register window? Explain working mechanism with suitable example. [See past questions & more numerals]
- (56) Explain Flynn's Taxonomy in detail.
- (57) Explain MIMD System Topology in detail.
- (58) Short notes:
- (a) Multicore architecture
 - (b) Microsequences
 - (c) RTL
 - (d) BCD
 - (e) Wallace tree
 - (f) DMA.
 - (g) UART
 - (h) Ring vs Mesh Topology
 - (i) Timing Diagram
 - (j) Use of microprocessor 8085
 - (k) Logical Instructions.
 - (l) Bus Interface Unit.
 - (m) CPU.